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5.R.R1 (1/1 point)

Download the file [5.R.RData \(/c4x/HumanitiesScience/StatLearning/asset/5.R.RData\)](/c4x/HumanitiesScience/StatLearning/asset/5.R.RData) and load it into R using `load("5.R.RData")`. Consider the linear regression model of y on X_1 and X_2 . To within 10%, what is the standard error for β_1 ?

Answer: 0.02593**EXPLANATION**

Use `summary(lm(y~.,data=Xy))`

[Hide Answer](#)*You have used 1 of 5 submissions*

5.R.R2 (1/1 point)

Next, plot the data using `matplot(Xy,type="l")`. Which of the following do you think is most likely given what you see?

- ☐ Our estimate of $s.e.(\hat{\beta}_1)$ is too high.
- ☒ Our estimate of $s.e.(\hat{\beta}_1)$ is too low. ✓
- ☐ Our estimate of $s.e.(\hat{\beta}_1)$ is about right.

EXPLANATION

There is very strong autocorrelation between consecutive rows of the data matrix. Roughly speaking, we have about 10-20 repeats of every data point, so the sample size is in effect much smaller than the number of rows (1000 in this case).

[Hide Answer](#)*You have used 1 of 5 submissions*

5.R.R3 (1/1 point)

Now, use the (standard) bootstrap to estimate $s.e.(\hat{\beta}_1)$. To within 10%, what do you get?

0.02883128

0.02883128

Answer: 0.0274

EXPLANATION

When we do the i.i.d. bootstrap, we are relying on the original sampling having been i.i.d. That is the same assumption that screwed us up when we used lm.

Hide Answer

You have used 1 of 5 submissions

5.R.R4 (1/1 point)

Finally, use the block bootstrap to estimate $s.e.(\hat{\beta}_1)$. Use blocks of size 100. To within 10%, what do you get?

0.1949591

0.1949591

Answer: 0.212

EXPLANATION

The block bootstrap does a better job of mimicking the original sampling procedure, because it preserves the autocorrelation.

Hide Answer

You have used 4 of 5 submissions

